

# Comparing Three Local LLMs for the NSE RAG Agent

*Llama 3.1 8B, Llama 3.2 3B, and Phi-3, evaluated on faithfulness, correctness, instruction adherence, and speed*

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## Purpose

The NSE RAG Agent uses a local LLM through Ollama, grounded in filings and news indexed in Chroma, to answer questions about NSE-listed companies. Two different Llama models had already been used in the project (3.1 8B and 3.2 3B), which raised an obvious question: was one actually better, or was the choice between them arbitrary? Phi-3 was added as a third model, so the comparison wasn't just two variants of the same family, and a small evaluation harness was built to score all three models on the same set of questions. The purpose of this study is to determine which of the three models is best suited to run this agent on-prem, not just by an overall average score, but by understanding where each model struggles. That distinction matters more for a financial assistant than a single headline number.

## Method

Twelve questions were written against data the agent had indexed (four tickers: RELIANCE, INFY, TCS, and HDFCBANK), split into three groups. Four questions asked about a single company, four asked something that spanned multiple companies, and four asked about tickers that don't exist. That last group was included deliberately, to test whether a model will admit it has nothing to go on or invent something that merely sounds plausible.

For each question, a reference answer was written based only on what the retrieval step surfaced, not on outside knowledge of these companies. This was a deliberate choice: the agent's system prompt instructs it to answer using only the provided context, so a model that stuck strictly to a thin set of retrieved chunks was behaving correctly, even where a richer answer existed in the real world.

Each of the 36 model responses (3 models across 12 questions) was scored by a separate judge model, ***Qwen2.5 7B***, on three dimensions: whether the answer was actually grounded in the retrieved context (faithfulness), whether it matched the substance of the reference answer (correctness), and whether it followed the system prompt's rules around avoiding investment advice and citing sources (instruction adherence). A model outside the comparison was used as judge so it wouldn't be grading its own family. Latency and generation speed were taken directly from Ollama's response metadata, without needing a judge.

## Results

Here's the overall picture across all 12 questions:

| Model        | Faithfulness | Correctness | Instr. adherence | Avg latency (s) | Tokens/sec |
|--------------|--------------|-------------|------------------|-----------------|------------|
| Llama 3.1 8B | 3.25         | 2.50        | 4.50             | 39.65           | 7.91       |
| Llama 3.2 3B | 3.42         | 2.92        | 5.00             | 17.96           | 17.32      |
| Phi-3        | 2.92         | 3.17        | 4.75             | 30.65           | 15.32      |

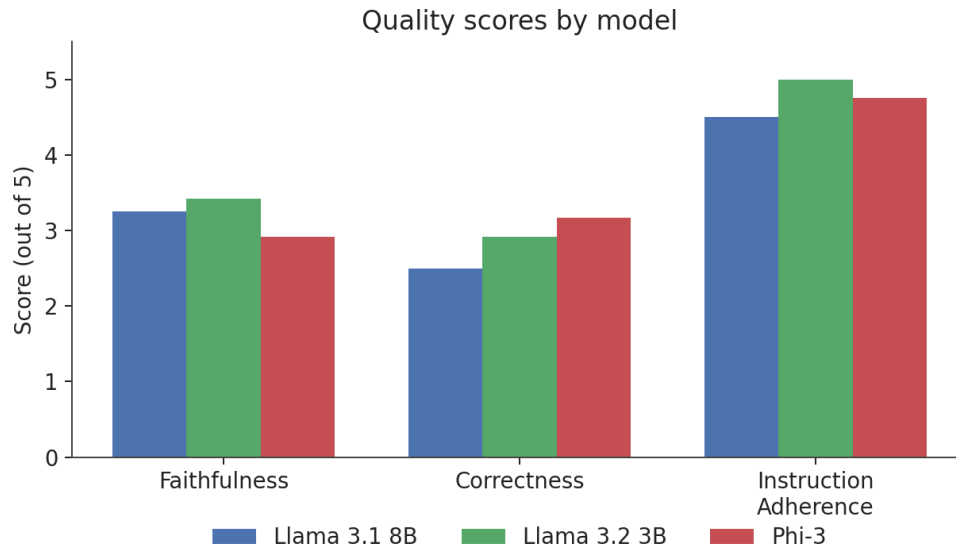


Figure 1. Quality scores by model, averaged across all 12 questions.

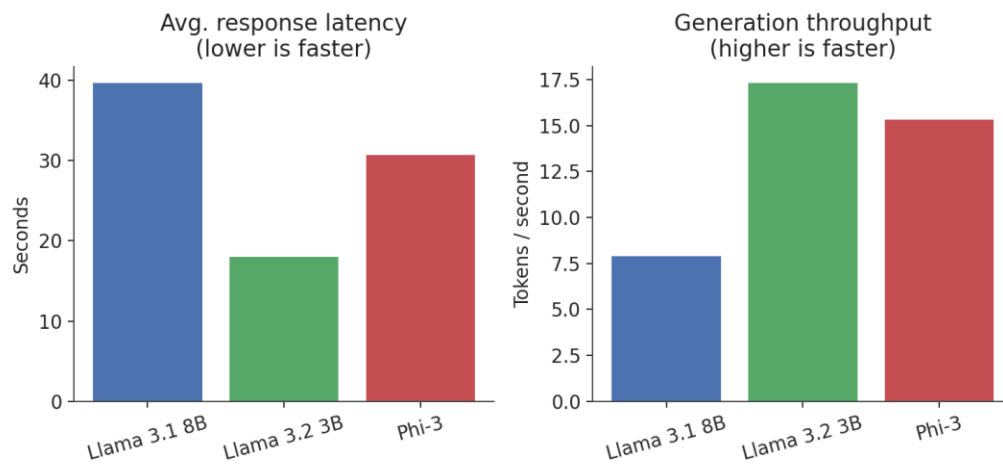


Figure 2. Response latency and generation throughput. Llama 3.2 3B was more than twice as fast as Llama 3.1 8B.

## What actually stood out

The overall averages are a reasonable summary, but the more useful finding emerges when results are split by question type instead of averaged together.

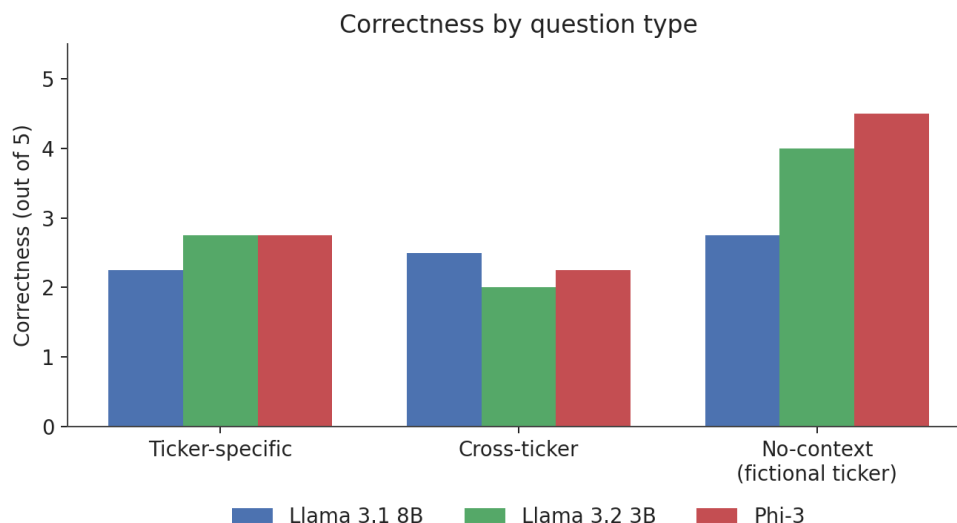


Figure 3. Correctness scores broken out by question category. The gap on 'no-context' questions is the main finding of this comparison.

Llama 3.1 8B, despite being the largest of the three models, was noticeably worse at handling the questions about fictional tickers. Its correctness score dropped to 2.75 there, while Llama 3.2 3B and Phi-3 held up much better at 4.00 and 4.50. A review of the transcripts shows a fairly clear pattern: Llama 3.1 8B tended to either invent a specific, confident-sounding detail (such as a filing that doesn't exist) or hedge vaguely rather than plainly stating it found nothing. The two smaller models were more likely to state the absence of information directly.

For a financial assistant, this is arguably the metric that matters most, more than any of the others measured here. A model that gives a slightly incomplete answer to a real question is a minor annoyance. A model that fabricates a filing when there's nothing to find is a much bigger problem, particularly in a domain where people might act on what it says.

Instruction adherence, by contrast, wasn't much of a differentiator. All three models stayed in the 4.25 to 5.00 range, consistently avoiding investment advice and generally citing which ticker or date they were drawing from. That part of the system prompt appears to hold regardless of which model sits behind it.

On speed, Llama 3.2 3B was the clear winner, and not by a small margin. It averaged 18 seconds per response against Phi-3's 31 and Llama 3.1 8B's 40 and produced more than twice the tokens per second of the largest model. Since it also led on faithfulness and instruction adherence, there's barely a trade-off involved in choosing it over Llama 3.1 8B. Phi-3 represents the one genuine trade-off in this comparison: it scored best on correctness but had the weakest faithfulness of the three, suggesting it's more willing to fill in gaps beyond what the context actually supports. Depending on the use case, that could be either useful or risky.

## Limitations

Twelve questions are enough to see a directional pattern, not enough to treat these numbers as statistically precise. A single question can shift a category average by a quarter point or more, so the overall pattern (the two smaller models outperforming the largest one, and the hallucination gap on unanswerable questions) should be read as the real finding, rather than the exact decimal scores.

The judge model has its own limitations as well. Qwen2.5 7B is a capable but imperfect judge, and only a handful of its scores were spot-checked against the actual transcripts rather than all 36. Further review of the judge's reasoning would be worthwhile before citing these numbers in a more formal setting.

Memory and VRAM usage per model were not measured. This was planned, but Ollama's process output didn't parse reliably in this setup, and resolving that didn't seem worth delaying the rest of the comparison for. As a result, the practical question of what hardware each model requires is only partly answered here, through latency and parameter count rather than a direct measurement.

Finally, all three models were tested against identical retrieval settings (same top-k, same embedding model), so this study isolates the generation model specifically. It does not test whether a different model might pair better with a different retrieval configuration.

## Recommendations

For this specific system, an on-prem RAG agent answering financial questions where fabricating unsupported facts is the costliest failure mode, Llama 3.2 3B is the strongest overall choice. It led on faithfulness and instruction adherence, held up second best against the fabrication problem, and was by far the fastest and cheapest to run. Phi-3 is a reasonable alternative where correctness on partially-covered questions matters more than strict grounding. Llama 3.1 8B, despite being the largest model tested, did not justify its slower speed with better accuracy or safety on this task.